
Context Rot Is Three Mechanisms: Only Displacement Acts Through Attention Mass

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Abstract

1 Long conversations are said to wear language models down. The phenomenology
2 is easy to elicit—as context fills, output entropy collapses several-fold and the last
3 token’s attention drifts off the system prompt and the current query onto recent
4 text—but the causal claim behind it has never been tested inside a single harness,
5 because the standard setups confound *accumulation* (more context) with *dilution*
6 (the answer-bearing tokens becoming a smaller, more distant fraction of it). We
7 separate them. Accumulating **individually localized** tasks, where the evidence for
8 the current answer always sits at the current query, we reproduce the full signature
9 battery across four model families and find **no accuracy cost** and no loss of instruc-
10 tion adherence. Holding items, model, fill and metric fixed and moving only *where*
11 *the evidence lives*, the cost appears at once: displacing it costs 0.19–0.21 accuracy,
12 and in a joint fit distance carries the coefficient ($\beta = -0.0076 [-0.0117, -0.0035]$)
13 while fill carries none. We then show this cost is **causally mediated by attention**
14 **mass**. Clamping the evidence’s attention share at its original position, with nothing
15 moved, reproduces **114%** of the displacement penalty (+0.202 [+0.138, +0.267],
16 landing indistinguishably on the displaced arm); a dose-response sweep over the
17 share recovers the same effect to within 0.004 and shows the mass→accuracy
18 curve is smooth and graded, with no threshold anywhere in it. The drain itself is
19 positional in *tokens*, not turns. Holding distance and fill fixed and varying only how
20 *confusable* the accumulated context is isolates a *second and separate* cost: near-
21 duplicate context is worth $-0.085 [-0.140, -0.030]$ while leaving the evidence’s
22 *total* attention mass unchanged ($-0.0003 [-0.0009, +0.0004]$, where the mea-
23 sured dose-response predicts 2% of the observed effect). That null is layer-specific:
24 measured across all 32 layers the same contrast removes 0.0022 of evidence mass
25 against displacement’s 0.0455, for accuracy costs of 0.085 and 0.188—cheap in
26 mass but not free, and the two drain profiles run in opposite directions across depth.
27 So displacement acts through attention mass and competition does not—and needle
28 benchmarks vary both at once, which is why they have not been told apart. A third
29 mechanism costs *compliance* rather than accuracy: accumulated filler erodes an
30 instructed output format only when it demonstrates an answer shape applicable to
31 the current probe—a drop-in shape collapses compliance $0.875 \rightarrow 0.000$ by fill
32 0.15, with accuracy unharmed, while off-domain filler holds compliance ≥ 0.88
33 through fill 0.78. That erosion is also not attention-mediated: the instruction’s
34 per-token attention is flat-to-rising in every arm, masking the demonstrations at
35 generation time restores nothing, and the instruction stays perfectly linearly de-
36 codable (AUC 1.000) while behaviorally inert. Yet attention mass is a causal
37 *weight* in the contest—clamping the instruction’s share down collapses compli-
38 ance ($0.99 \rightarrow 0.03$) where nothing in context competes, clamping it up restores it
39 against 42 counter-exemplars, and restating the policy restores it without surgery.
40 Meanwhile the signatures are in-context comfort: the entropy collapse vanishes
41 on organic WildChat dialogue and tracks conversation homogeneity, not length,

Table 1: **Distance is the variable that matters** (OLMo-2-Instruct, $n=192$ per arm, chance = 0.200, fill 0.688 in every arm). Evidence attention share is measured at layer 24 on the same forwards. Case-resampled bootstrap 95% CIs.

arm	local	back_2	back_5	back_10	back_20
accuracy	0.464	0.359	0.292	0.250	0.276
cost vs local	—	+0.104	+0.172	+0.214	+0.188
95% CI	—	[+.005, +.203]	[+.073, +.266]	[+.120, +.307]	[+.094, +.281]
evidence share @L24	0.0408	0.0363	0.0320	0.0195	0.0124

and across OLMo-2’s post-training chain it is a baseline-confidence level shift installed mostly by SFT. Context rot, as measured, is not a decay of competence with length: accumulation alone is free; what costs accuracy is the evidence being displaced or contested, and what costs compliance is the model’s own applicable precedent—and only displacement is attention dilution.

1 Introduction

A recurring worry about long conversations is *context rot*: as turns accumulate, the model is said to degrade. The phenomenology is real. On a repeating multiple-choice task, instruction-tuned models become dramatically more confident as context fills—next-token entropy drops 3–4 $\times$ on Qwen-2.5-7B and Llama-3.1-8B, in both question orders—and the last token’s attention shifts systematically away from the system prompt and the current query toward the most recent prior cases, with the distribution diffusing (at layer 24 of OLMo-2-Instruct: system $\leftrightarrow$ fill -0.93 , recent $\leftrightarrow$ fill $+0.89$, current $\leftrightarrow$ fill -0.71 , entropy $\leftrightarrow$ fill $+0.95$, all monotone in fill). By every visual diagnostic this is a strong effect that grows with context.

The tempting reading is that the model is being worn down by length. But almost every context-rot evaluation embeds the answer-bearing information inside *one long task*—a long document, a buried needle, a growing codebase. There, longer context mechanically *dilutes* the relevant tokens among irrelevant ones, so a measured degradation conflates the cost of accumulation per se with the cost of the target being harder to attend to (“lost in the middle,” Liu et al., 2023). These cannot be told apart in that setup, and the distinction matters practically: an agent transcript of finished, locally-scoped steps accumulates without diluting, and the two accounts make opposite predictions about it.

In this paper we separate them inside one harness. We accumulate **individually localized** tasks, so that context length and the dilution of the answer-bearing evidence become independent knobs, and we intervene on each in turn. Accumulation alone costs nothing. Moving the evidence k turns back, at identical fill and byte-identical questions, costs 0.19–0.21 accuracy (Table 1).

That cost is *causally mediated by attention mass*. A clamp that sets a span’s post-softmax attention share lets us remove the evidence’s mass while leaving it exactly where it is: doing so reproduces 114% of the displacement penalty, and a dose-response sweep over the share recovers the same endpoint to within 0.004 across two separately-run experiments.

A second cost is not mediated that way at all. Holding the evidence at the current query and varying only how *confusable* the accumulated context is, near-duplicate context costs a further 0.085 accuracy while the evidence’s attention share does not move measurably—2% of what the measured dose-response would require.

A third cost falls on *compliance* rather than accuracy. Holding the evidence local and accumulating filler behind a format instruction, the instructed output format survives off-domain filler through fill 0.78 but collapses from 0.875 to 0.000 by fill 0.15—three turns—of filler whose answers demonstrate a shape applicable to the probe, never to recover, while accuracy on the same replies is untouched. The instruction’s per-token attention is flat-to-rising through the collapse, and the instruction remains perfectly decodable from the residual stream while it has zero behavioral force; yet clamping its attention down collapses compliance where nothing in context competes, and either clamping it up or restating the policy fully restores compliance against forty-two counter-exemplars. Format erosion is a *precedent contest* in which attention mass is a causal weight but not the natural driver.

84 Taken together, context rot is three mechanisms with different signatures, and a needle-in-a-haystack
85 benchmark varies the first two at once, which is why they have not been told apart.

86 Our contributions are as follows:

- 87 • A harness that makes accumulation and dilution independent, by accumulating individually
88 localized tasks whose evidence never moves. In it, accumulation alone costs no accuracy and no
89 instruction adherence, with bounded nulls rather than point estimates.
- 90 • A causal mediation result for displacement: removing the evidence’s attention mass with nothing
91 moved reproduces 114% of the displacement penalty, and the mass→accuracy relationship is
92 smooth and graded with no threshold in it.
- 93 • A dissociation establishing a *second* mechanism: competition costs accuracy at constant attention
94 mass, so “attention dilution” is the right account of displacement and the wrong account of
95 competition.
- 96 • A *third* mechanism at the compliance level: accumulated demonstrations erode an instructed
97 format only when their shape is applicable, immediately and with accuracy unharmed. The
98 erosion is not attention-mediated—but the instruction’s attention mass is a causal weight in the
99 underlying contest, and re-weighting or re-presenting the instruction each fully reverse it.

100 2 Background

101 **Length as the suspected cause.** Most evidence for context rot comes from setups where the
102 answer-bearing information sits inside a single long input. Liu et al. (2023) showed that accuracy
103 depends strongly on *where* in the context the relevant passage sits, dipping when it falls in the middle.
104 Levy et al. (2024) held a reasoning task fixed and padded the input with task-irrelevant text, finding
105 degradation well before the models’ advertised context limits. Hong et al. (2025) ran a broad sweep
106 over 18 models and found that reliability falls with input length even on tasks as simple as retrieval
107 and replication, and that the fall is not uniform: it depends on how similar the needle is to the question,
108 and on whether distractors are present. Laban et al. (2025) report a large multi-turn degradation, but
109 locate its cause in the model committing early to a wrong reading and failing to recover—not in
110 length as such.

111 **Confusable context as a separate suspect.** A second literature manipulates the *content* of the
112 surrounding context rather than its length. Shi et al. (2023) insert an irrelevant sentence into
113 grade-school math problems and observe a sharp accuracy drop. In retrieval-augmented generation,
114 Cuconasu et al. (2024) find that documents retrieved as relevant but not containing the answer act as
115 distractors and degrade accuracy.

116 **Instructions versus demonstrations.** Demonstrations can override a model’s trained priors (Wei
117 et al., 2023) and, at scale, its safety training (Anil et al., 2024); hardcoded assistant turns exhibiting a
118 competing pattern flip instruction-following into pattern-following (Camassa and Shiller, 2026), and
119 instruction-derived and demonstration-derived task representations are measurably distinct (Davidson
120 et al., 2025). Closest to our question, Dongre et al. (2026) track attention from responses to the
121 system prompt declining over 50 filler turns and show that hard-masking the instruction out of the
122 window collapses compliance. In each case the instruction’s attention is measured alongside the
123 behavioral change or ablated outright, never *set*; and the eroding demonstrations are supplied by the
124 experimenter rather than accumulated from the model’s own benign replies.

125 **What is not separated.** In each of these setups the two candidate causes move together. Padding a
126 prompt lengthens it *and* pushes the evidence further from the query; adding a distractor document
127 does the same while also supplying a competing answer. So a measured degradation cannot be
128 assigned to accumulation, to the evidence becoming distant, or to the surrounding text becoming
129 confusable. A mechanistic thread—Liu et al. (2023)’s positional account, and attention-based
130 explanations generally—suggests attention as the channel, but attention is measured alongside these
131 manipulations rather than intervened on. This paper separates these factors inside one harness and
132 intervenes on attention directly.

3 Preliminaries

Localized task streams. Our harness accumulates **individually localized tasks**: a stream of self-contained 5-option medical cases (DDXPlus, Tchango et al., 2022) or discrete factual questions (MMLU, Hendrycks et al., 2021). Each answer depends only on the *current* question, which always sits at the end of the context; prior turns are never required to answer it. Accumulation therefore grows the context without ever diluting the information the current answer needs. A reversed-question-order control removes difficulty-ordering confounds. Because a 4–32k window holds only tens of cases, we pool many independent accumulation sessions.

The dilution knob. The design’s real value is that dilution becomes an *independent knob*. Holding the item, the model, the total fill and the metric fixed, we can move the evidence k user turns back into the accumulated transcript, with the question text byte-identical across arms and the same filler in every arm. Distance and fill, welded together in a long-document setup, move independently here. All causal experiments run on OLMo-2-7B-Instruct at layer 24, seed 42, with an overflow guard that skips rather than truncates every item that would not fit whole—truncating a long item near the window edge manufactures exactly the late-window cost under study.

The attention-mass clamp. To intervene on attention rather than position, we add a clamp that *sets* a token span’s post-softmax attention share: a constant bias on the span’s key columns scales its odds, and softmax renormalizes, so nothing in the attention forward is reimplemented and a scale of 1.0 is bit-identical to the unhooked model. The clamp is uniform across heads.

Analysis. Per-condition n is bounded by the context window, so every clamp experiment is paired over shared items and analysed with a paired bootstrap resampling *cases* (10,000 draws), never heads or turns.

4 Results

4.1 Accumulation

Per-case accuracy does not fall as context fills; it is worst at the cold start and warms up ($\text{corr}(\text{correct}, \text{fill}) = +0.11$ Instruct, $+0.07$ DPO). The obvious objection is that a homogeneous repeating task lets in-context learning mask a hidden decay, so we remove that crutch: in a *random-subject* stream each question is drawn uniformly from all 56 MMLU subjects, so prior context carries no predictive information about the next question—only the answer *format* is learnable. Accuracy is still flat (random: $\text{corr} = -0.02$ $[-0.10, +0.05]$, $n=699$; coherent single-subject: $\text{corr} = +0.01$, $n=1001$). These nulls are *bounded*: the 95% interval on upper-minus-lower-half accuracy excludes declines beyond 4.1 points (coherent) and 9.4 (random). Checkable instruction adherence—a canary system instruction to begin every reply with a marker and end with a tag—stays at ceiling throughout ($\text{corr}(\text{violation}, \text{fill}) = 0$), an honest null that bounds rather than proves, since the canaries were easy; §4.4 shows what it does and does not mean.

4.2 Displacement

We now vary only *where the answer-bearing evidence lives*. Each DDXPlus probe’s vignette is placed k user turns before its question; mean fill is 0.688 in every arm, the question text is byte-identical, and the overflow guard skipped 0 of 192 probes. Every gap’s CI excludes zero (Table 1, Fig. 1a). In a joint fit of accuracy on fill and distance, **distance carries the coefficient** ($\beta = -0.00761$ $[-0.01173, -0.00346]$, significant) and **fill carries none** ($\beta = -0.00725$ $[-0.21006, +0.18658]$). Within the *local* arm—the one that reproduces the paper’s original design—accuracy is flat with fill ($\beta = -0.294$ $[-0.767, +0.184]$). The effect survives restriction to parsed responses and is then fully monotone ($0.524 \rightarrow 0.413 \rightarrow 0.397 \rightarrow 0.343 \rightarrow 0.338$). One honest qualification: the decline saturates by $k \approx 10$ rather than deepening to $k=20$, and those two arms are not distinguishable from each other. So the same model, on the same items at the same fill, degrades when the evidence is displaced and does not degrade when it is not. The localized null is not a ceiling artifact: these items *can* show degradation, and do, as soon as dilution is reintroduced.

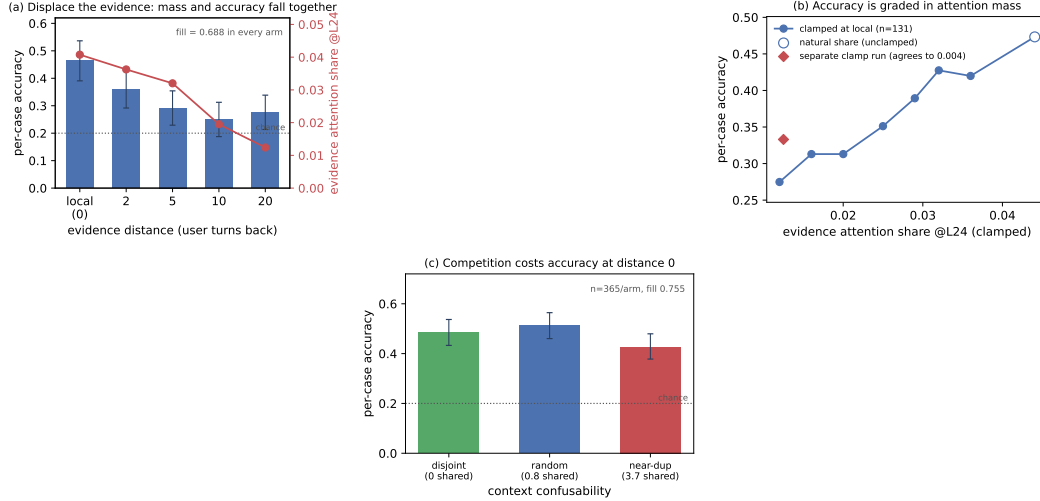


Figure 1: **Dilution costs accuracy; accumulation does not.** (a) Moving the evidence k user turns back, at identical fill and byte-identical questions, drains its attention mass and costs accuracy together. (b) Setting that mass directly, with the evidence left in place, reproduces the penalty and traces a smooth graded dose-response—no threshold. The diamond is an independently-run clamp experiment measuring the same endpoint cut; the two agree to 0.004. (c) At distance 0 and fixed fill, near-duplicate accumulated context costs a further 0.085. Bootstrap CIs resample cases throughout.

Distance and attention drain move together by construction, so Table 1 is consistent with dilution but does not establish it—a purely positional account predicts the same table. Four interventions separate them.

Observational correlations. Evidence share falls $0.0408 \rightarrow 0.0124$ across the ladder ($r = -0.83$ with distance) while the *question*’s share stays flat, confirming only the evidence moved. A trap worth recording: *within* an arm, higher evidence share predicts *lower* accuracy ($\beta = -11.2 [-20.0, -2.6]$, stronger controlling for vignette length). Attention there tracks item difficulty—the model looks harder at confusing vignettes—and has the *opposite sign* to the causal effect below. Anyone reading share-against-accuracy correlations off this data without intervening will conclude the reverse of the truth.

Mass removal. Keeping the evidence at local and clamping its share down to the *same item*’s own back_20 share costs $+0.202 [+0.138, +0.267]$ (paired $n=174$) and lands at 0.333 against back_20’s 0.365—a difference of $-0.025 [-0.083, +0.035]$, indistinguishable. **Mass removal alone reproduces 114% of the displacement penalty, with nothing moved.**

Dose-response. Sweeping the evidence’s share at fixed position over seven levels (common subset $n=131$ present at every level) gives a monotone decline from 0.473 at the natural share (0.0441) to 0.275 at 0.012 (Fig. 1b). Six of the seven levels differ significantly from the natural share—including a cut of less than a fifth of the mass—and no step is a threshold: the largest adjacent step is 0.053 and the decline is smooth throughout. The endpoint contrast, $+0.198 [+0.115, +0.282]$, matches the independently-run clamp experiment’s $+0.202 [+0.138, +0.267]$ for the same share change—agreement to 0.004 across two separate runs, which is the strongest internal consistency check in the program. An earlier draft of this work read a *threshold* into two flat contrasts; the sweep refutes it. The relationship is *graded*: a measured slope, not a cliff.

Mass restoration. The converse is only partial: clamping the displaced evidence’s share back *up* to its local level—share-matched across all 32 layers, not indexed to one—recovers a real but small 0.28 $[0.07, 0.61]$ of the penalty ($+0.047 [+0.010, +0.083]$ of $+0.167 [+0.099, +0.240]$), leaving most of it in place. Our uniform clamp can strip mass faithfully but cannot reconstruct *which* heads should carry the restored mass. Full sufficiency with partial necessity is the signature of an instrument that is faithful in one direction and crude in the other, and we report it as one rather than as a second mechanism. A pattern-matched, per-head clamp is the test that would settle it.

Tokens versus turns. A partial 2×2 in (gap turns) $\times$ (filler length), total context equalized by padding, dissociates them: two arms matched on *tokens* have effectively identical evidence share (0.0104 vs 0.0108) despite a $4 \times$ difference in turn count, while at matched turns share falls $0.0288 \rightarrow 0.0104$ as the gap grows $446 \rightarrow 1684$ tokens—a 64% cut in the evidence’s attention that costs at most 9.4 accuracy points ($+0.026 [-0.042, +0.094]$). Share regresses significantly on gap tokens and not on gap turns. Attention drain is a function of token distance, as RoPE decay predicts; conversation structure is not what drains it.

The current query. Accumulation drives the *current query*’s share from ≈ 0.35 down to ≈ 0.15 , and it is natural to ask whether that stops short of a floor. It does not. Clamping the query span on cold-start contexts, accuracy is flat from the natural share (0.258, accuracy 0.545) down through 0.20 (0.509), then costs 16.4 points at 0.15 ($+0.164 [+0.036, +0.291]$)—which is exactly the share accumulation reaches. Accumulation does not stop far short of the shoulder; it stops *at* it. (Levels below 0.15 require -4.7 to -6.1 nats, near-ablation rather than points on a dose-response—the model answers “A” 110/110—and are excluded from interpretation.) That accuracy nonetheless stays flat under accumulation, while the same share reached by clamping costs points, is itself evidence that in-context learning is actively holding accuracy up.

4.3 Competition

Burying a needle varies two things at once: the evidence becomes distant *and* it becomes surrounded by plausible alternatives. So far we have isolated distance. We isolate the other half by holding the evidence at the current query and fill fixed, and varying only how *confusable* the accumulated context is. Every arm accumulates eight prior DDXPlus cases, each answered in the transcript with its gold letter, so format and in-context-learning affordance are constant; the arms differ only in how much the accumulated cases’ differentials overlap the current probe’s (disjoint 0 shared options, random 0.80, near_dup 3.75 of 5, always with a *different* gold, so the context never displays the probe’s answer as a correct one). Paired over $n=365$ probes, accuracy is random 0.512, disjoint 0.485, near_dup 0.427.

Near-duplicate context costs $-0.085 [-0.140, -0.030]$ against the natural stream and $-0.058 [-0.112, -0.006]$ against disjoint context; in a joint fit within this experiment, **option overlap carries the coefficient** ($\beta = -0.021 [-0.039, -0.003]$) and **fill carries none** ($\beta = -0.284 [-0.640, +0.075]$)—the same shape as the distance result, on a different variable. Two controls matter: the low-competition arms must land on §4.2’s local accuracy of 0.464, and both do ($+0.049 [-0.037, +0.134]$ and $+0.021 [-0.065, +0.107]$); and context length does not order the arms, since disjoint is the *longest* and mid-accuracy while near_dup is the shortest, so length biases against the result. Restricted to parsed responses only, random – near_dup = $+0.082 [+0.025, +0.138]$ survives. The effect is *not* monotone in confusability: random sits above disjoint, not below it, though not significantly ($+0.027 [-0.019, +0.074]$). Mild topical overlap plausibly supplies useful pruning over the option space while near-duplication supplies competitors that actively mislead; only the second is large enough to measure here. We therefore claim that near-duplicate context is costly, *not* that cost rises with overlap—one significant step, not a gradient.

Attention mass under competition. Repeating the sweep with attention capture instead of generation (same probes, same seeds), and measuring every one of the model’s 32 layers rather than a reference layer, separates the two mechanisms on mass. Averaged over all layers, displacement removes 0.0455 of the evidence’s attention share and competition removes 0.0022, while costing 0.188 and 0.085 accuracy respectively. Competition is therefore far cheaper in mass per point of accuracy than displacement, which is the dissociation—but it is *not* mass-neutral, and the layer chosen decides how it looks. At layer 24 the headline contrast moves the evidence’s share by $-0.00027 [-0.00088, +0.00035]$, indistinguishable from zero; at layer 17 it moves it by -0.0186 , and at layers 16 and 18 by -0.0110 and -0.0093 . The two drain profiles are moreover *negatively* correlated across layers ($r = -0.32$): displacement bites hardest at layers 0–3 and 16, competition at 17–18. That the same accuracy contrast has opposite attention signatures in different layers is evidence for two mechanisms which no single-layer measurement could supply. An earlier version of this work reported that reproducing competition’s cost through mass would need a share change $50 \times$ larger; that figure was read off layer 24 alone and does not hold across the model.

Per head. Attention share is a mean over a layer’s 32 heads, and a mean can hold still while heads cancel, so we also measured all 1,024 heads. Under displacement at layer 24, **every head loses evidence mass** from `local` to `back_20`, each significant at Bonferroni, so nothing is hidden by averaging. The loss is close to uniform in proportion—mean fractional drain 0.689 (sd 0.147), uncorrelated with how much mass a head held ($r=0.08$)—and a single uniform odds-scale reproduces 58% of the per-head pattern, which weakens (without eliminating) the reading of §4.2’s partial necessity as pure instrument limitation. Under competition at the same layer the net is 0.0003 while heads move 0.0026 against a paired sign-flip null of 0.0004 ($p \leq 0.0005$, 2,000 permutations), with 19 of 32 heads individually significant: a small reallocation, about 6% of the evidence’s local share, rather than inaction. Finally, heads that concentrate on the evidence do exist—255 of 1,024 attend it more than its token share warrants, up to 0.626 for one head on a span that is 0.102 of the context—but none of them are at layer 24, whose mean evidence share (0.0408) is among the lowest in the model. Conclusions about head specialisation drawn from a single layer do not generalise, and we draw none.

4.4 Precedent

The costs so far are to accuracy. A third mechanism costs *compliance*: not whether the model can still solve the task, but whether it answers in the instructed form. The system prompt carries a clinical answer template (ANSWER: letter, then SUPPORTING: with at least two vignette findings), the context accumulates the model’s own replies to interleaved filler, and every probe reply is graded separately for compliance and accuracy ($n=40$ probes per depth, one accumulation snapshot at every depth, so depth is the only mover).

The instruction’s attention is causally necessary—and demonstration masks it. Accumulation drives the system span’s share down $8\times$ within eight turns ($0.166 \rightarrow 0.021$). Clamping it from 0.165 to 0.050—a level accumulation passes on its own between two and four turns, at -2.6 nats, well clear of the near-ablation regime excluded in §4.2—in a context containing no assistant turn collapses compliance from 0.99 to 0.03 (a suffix canary flips on 120 of 120 items) while accuracy moves $0.525 \rightarrow 0.467$ ($[-0.017, +0.133]$) at parse rate 0.967: compliance is destroyed and competence is not. But one *compliant* example in context pins compliance at ceiling at every clamp level, and one bare-letter example pins it at floor even unclamped—replies byte-copy the previous assistant turn’s format, with zero length variance across 720 generations per arm. The instruction governs behaviour only when the transcript is silent, which re-reads §4.1’s adherence null: that null accumulates *compliant* history, so it shows compliance survives compliant precedent, not accumulation.

Erosion tracks applicability, not attention. Three filler streams differ in what their answers demonstrate: `code` (request→Python function; no shape applicable to a 5-option probe), `gsm8k` (worked prose; loosely applicable), `mm1u` (MCQ→bare letter; a drop-in replacement). `code` never erodes (compliance 0.875 at cold start, 1.00 at matched fill 0.5, 0.90 at fill 0.78); `gsm8k` erodes late and partially (0.600 at fill 0.48, replies drifting into the filler’s worked-solution register); `mm1u` collapses totally and immediately, $0.875 \rightarrow 0.000$ at three filler turns (fill 0.147), never recovering through fill 0.87. At matched fill ≈ 0.5 the arms order 1.00/0.60/0.00. Accuracy on the same replies is unharmed everywhere (0.425 shallow, 0.500–0.684 deep): the model still solves the case—it answers in the demonstrated style. The system prompt’s attention does not carry the difference: its raw share falls on the same arithmetic curve in every arm, its per-token enrichment is flat-to-rising in all three ($1.51 \rightarrow 3.07$ in `code`, $1.50 \rightarrow 2.14 \rightarrow 1.47$ in `mm1u`), and `code` holds full compliance at half the raw share at which `mm1u` has already fully collapsed.

The mode is installed at prefill, and linear probes read its shadow. The demonstrated answer spans *are* preferentially read exactly where erosion occurs (per-token enrichment of filler answers over filler questions: +1.28 in `mm1u`, +0.34 in `gsm8k`, negative in `code`—the erosion ordering), but reading them is not what sustains it: masking every exemplar answer’s key columns at generation time restores nothing (0.000 compliant; a size-matched random-masking control is equally null), so the privileged reading is an epiphenomenon of the template’s installation (Hendel et al., 2023). Residual-stream probes complete a three-level dissociation: the instruction’s presence remains perfectly linearly decodable at every depth (transfer AUC 1.000, permutation $p=0.000$) while it has zero behavioral force—read, represented, overruled—and whether the upcoming reply will comply is decodable *before the first token* (leave-one-out AUC 0.822 at layer 21) within cells matched on depth, fill and

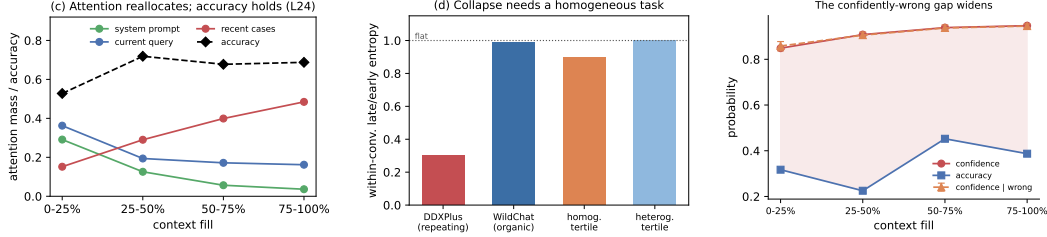


Figure 2: **The signatures are in-context comfort, and the residual hazard is calibration.** (a) Attention reallocates monotonically off the system prompt and current query with fill while accuracy holds. (b) The entropy collapse appears only with a homogeneous repeating task, not on organic WildChat dialogue, and tracks homogeneity rather than length. (c) Confidence rises with fill on wrong answers as much as right, at flat accuracy.

filler. The mode is installable by direction: adding the mean-difference vector at decode time to a demonstration-free context destroys the instructed format (0.000 vs. 0.525 under a norm-matched random vector), the replies reproducing the demonstrated registers verbatim. But no linear direction we tried *erases* it—mean-difference at one layer and at eleven, and the probe’s own reader direction, all with clean controls—even though two fitted projections drive the layer’s linear code to chance: the code is low-rank, not redundant, so the behavioral decision does not consume the linearly decodable component. Decodability claims about instruction “presence” should be read accordingly.

Reversal is total, by either lever. At depth 42 (natural compliance 0.000, accuracy 0.675), clamping the system span back to its cold-start share restores compliance to 1.000 at an accuracy cost of $0.675 \rightarrow 0.425$ —mass forced onto a 48-token span is taken from everything else, §4.2’s zero-sum arithmetic—and restating the policy in the latest user turn also restores 1.000, at 0.500 and with no surgery (both at once: 1.000 at 0.275). Natural dilution therefore does not cause the erosion—it is identical in the arm that never erodes; an applicable competing template does—and the outcome is a contest in which attention mass is a causal *weight*: clamp the instruction down and it loses unopposed, clamp it up and it wins against 42 counter-exemplars, leave it natural and the demonstrated template beats it. Re-presenting beats re-weighting, though both work.

4.5 Confidence signatures

If the entropy collapse were a generic depth effect it should survive on heterogeneous dialogue. It does not. On 150 organic WildChat conversations (Zhao et al., 2024) (Qwen-2.5-7B, teacher-forced over the real history), within-conversation entropy is flat with depth: median late/early ratio 0.99, median within-conversation $\text{corr}(\text{entropy}, \text{depth}) -0.02$, and 52% of conversations trend down—a coin flip. Partitioning WildChat by its *own* inter-turn homogeneity isolates the driver: the entropy slope correlates with homogeneity (partial $r = -0.151$ controlling for length), homogeneous conversations collapse (0.897 late/early) while heterogeneous ones are flat (1.001), and because homogeneous conversations are *longer*, length works against the effect.

Replaying one probe across OLMo-2-7B’s (OLMo Team, 2024) $\text{base} \rightarrow \text{SFT} \rightarrow \text{DP0} \rightarrow \text{Instruct}$ chain (identical cases and orders) shows early-context entropy collapsing monotonically, $1.06 \rightarrow 0.58 \rightarrow 0.33 \rightarrow 0.20$, with the largest single drop at $\text{base} \rightarrow \text{SFT}$. But the “collapses as context fills” framing is mostly a base-model ICL effect plus a floor: the within-context ratio *falls* across stages ($1.64 \rightarrow 0.47$), because aligned models start near a confidence floor. Post-training is a downward *level shift* in baseline entropy, not a steeper collapse slope. Finally, the mechanistic story proposed for the collapse does not survive a causal test: clamping the boundary-detector SAE feature (Gemma Scope F90871, Lieberum et al., 2024) back to its clean level on held-out fatigued cases moves entropy the *wrong* way ($0.442 \rightarrow 0.465$ vs. clean 0.293), leaves accuracy flat, and a random feature of similar magnitude perturbs entropy at least as much.

Consistently, conditioning on fill, the cases OLMo-2 answers *wrong* have *higher* current-query attention than the ones it gets right (pooled $\Delta = +0.045 [-0.003, +0.097]$, $n=115$; sign positive in every fill bin and at every probed layer). Errors are associated with *fixating* on the query; correct answers spread attention onto the accumulated cases—the per-case fingerprint of the in-context

learning that keeps accuracy up. The “rot” and the within-context predictor of a correct answer point in opposite directions.

5 Conclusion and Outlook

Limitations. Per-condition n is bounded by the context window, so every interval here depends on the paired analysis of §3; resampling the arms independently inflates each one about $2.5\times$, enough to report a real effect as null. The necessity direction of the mediation is only partially established and is plausibly instrument-limited, since the clamp is uniform across heads. The competition result is one significant step rather than a gradient—mild overlap is indistinguishable from none, and we do not claim cost rises with confusability. The instruction-canary null is a floor effect, the WildChat figures are summary values from teacher-forced analyses, and the causal program runs on one model family. The precedent results are one instruction family, graded by a rule-based checker at $n=40$ per cell; and the carrier of the demonstrated-format mode is unidentified— it is decodable and installable, but no linear direction we tried erases it, so we localize the decision neither to a layer nor to a direction. The accuracy null holds for localized tasks; genuinely length-dependent single-task settings are where dilution is the point, and are out of scope by design.

With accumulation, displacement, competition and precedent separated inside one harness: accumulation produces a real, monotone reallocation of attention and a real collapse of *confidence*, but no loss of accuracy and no loss of instruction adherence. Displacing the evidence produces a large accuracy cost that is causally mediated by its attention mass, graded in that mass, and positional in tokens. Making the context confusable produces a further cost that is *not* mediated by that mass at all. Accumulating an applicable demonstrated answer shape erodes not accuracy but *compliance*, in a precedent contest the instruction loses with its attention intact—and which either re-weighting or re-presenting the instruction wins back completely. The literature’s central claim is therefore *correct but misattributed, and under-counted*: context rot is not a decay with length, it is displacement, competition and precedent, three mechanisms with different signatures, of which length is merely the usual way of causing them at once.

The residual hazard in the localized case is *calibration*, not competence: on the DDXPlus streams logging per-token confidence ($n=154$ pooled cases, two models, both question orders), answer confidence rises $0.85 \rightarrow 0.95$ with fill ($r = +0.72 [+0.64, +0.79]$) at flat accuracy—and equally on *wrong* answers ($0.86 \rightarrow 0.95$; $r = +0.69 [+0.57, +0.78]$)—so the confidently-wrong gap widens by ~ 10 points over a session (Fig. 2c), most consequential exactly where a homogeneous high-stakes stream makes the model surest. The test we would run next is the pattern-matched, per-head clamp that would settle whether the partial necessity above is an instrument limitation or a second channel within displacement itself.

References

- C. Anil et al. Many-shot jailbreaking. *NeurIPS*, 2024.
- A. Camassa and B. Shiller. Do as I say, not as I do: instruction-induction conflict in LLMs. *arXiv:2605.20382*, 2026.
- F. Cuconasu et al. The power of noise: redefining retrieval for RAG systems. *SIGIR*, 2024.
- G. Davidson, T. Gureckis, B. Lake, and A. Williams. Do different prompting methods yield a common task representation in language models? *NeurIPS*, 2025.
- V. Dongre et al. When attention closes: how LLMs lose the thread in multi-turn interaction. *arXiv:2605.12922*, 2026.
- R. Hendel, M. Geva, and A. Globerson. In-context learning creates task vectors. *Findings of EMNLP*, 2023.
- D. Hendrycks et al. Measuring massive multitask language understanding. *ICLR*, 2021.
- K. Hong, A. Troynikov, and J. Huber. Context rot: how increasing input tokens impacts LLM performance. Chroma technical report, 2025.
- P. Laban, H. Hayashi, Y. Zhou, and J. Neville. LLMs get lost in multi-turn conversation. *arXiv:2505.06120*, 2025.

- 406 M. Levy, A. Jacoby, and Y. Goldberg. Same task, more tokens: the impact of input length on the reasoning
407 performance of large language models. *ACL*, 2024.
- 408 N. F. Liu et al. Lost in the middle: how language models use long contexts. *TACL*, 2023.
- 409 T. Lieberum et al. Gemma Scope: open sparse autoencoders everywhere all at once on Gemma 2. 2024.
- 410 OLMo Team. OLMo 2: open language models with a full post-training recipe. 2024.
- 411 F. Shi et al. Large language models can be easily distracted by irrelevant context. *ICML*, 2023.
- 412 A. Fansi Tchango et al. DDXPlus: a new dataset for automatic medical diagnosis. *NeurIPS Datasets and*
413 *Benchmarks*, 2022.
- 414 J. Wei et al. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.
- 415 W. Zhao et al. WildChat: 1M ChatGPT interaction logs in the wild. *ICLR*, 2024.